

EE1060: Differential Equations and Transform Techniques**Practice Problem Set - 02**

Practice problems in topics covered till: Lecture 06

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Derive the current response of a series RL circuit excited by a DC voltage source V_{dc} . Assume that the voltage source is applied to the circuit by closing a switch at the instant $t = 0$, at which time the current through the inductor is I_0 .

2. Solve the first order ODE

$$\frac{dy}{dx} + ky = A \cos \omega x \quad (1)$$

where k is a constant. Derive a particular solution for the initial condition $y(0) = Y_0$.

3. Solve the first-order ODE

$$(x + xy^2) dy + (3x^3 + 2x) dx = 0 \quad (2)$$

4. Solve the first-order ODE

$$(xy + y) dx + \left(\frac{1}{2}x^2 + \frac{3}{4}y^2 \right) dy = 0 \quad (3)$$

5. Solve the first-order ODE

$$(2x + y^2) dx + (3xy + y) dy = 0 \quad (4)$$

6. Solve the first-order ODE

$$\frac{dy}{dx} = -\frac{y}{x^2 + y^2} \quad (5)$$

7. Solve the first-order non-linear ODE

$$\frac{dy}{dx} + ay = by^2 \quad (6)$$

where a and b are constants.

8. Solve the first-order non-linear ODE

$$x dy - (2x^2 + 4y) dx = 0 \quad (7)$$

9. Solve the first-order ODE

$$\frac{dy}{dx} + \frac{y}{x} = \ln(x) \quad (8)$$

10. Let $y(t)$ represent the price of the stock at time t . The rate of change of the stock price is assumed to be proportional to both the current price and the remaining potential growth in the market, subject to diminishing returns as the stock price approaches a market saturation level, S , which represents the maximum price the stock can realistically achieve. The rate of change of the stock price is described by the following differential equation:

$$\frac{dy}{dt} = ry \left(1 - \frac{y}{S}\right)^n \quad (9)$$

where, r is the growth rate constant (assumed to be 0.05), S is the saturation price or carrying capacity of the market (given as 1000), n is a constant that characterizes the degree of diminishing returns (assumed to be 2). At time $t = 0$, the initial stock price is given by $Y(0) = 200$. Determine $y(t)$ for the given initial conditions.

Hint: Use the following substitution: $u = \left(1 - \frac{y}{S}\right)$.

11. (Taken from [1]) Solve the first order ODE

$$\frac{4y^2 - 2x^2}{4xy^2 - x^3} dx + \frac{8y^2 - x^2}{4y^3 - x^2y} dy \quad (10)$$

12. (Taken from [1]) Determine the value of n for which the following first-order ODE is exact.

$$(x + ye^{2xy}) + nxe^{2xy} dy = 0 \quad (11)$$

13. Check if $u(x, y) = xy$ is an integrating factor for the following differential equation

$$\left(\frac{1}{2}xy - \frac{1}{x}\right) dx + \left(\frac{1}{3}x^2 - \frac{1}{y}\right) dy = 0 \quad (12)$$

If it is, find the general solution of the equation.

14. (Taken from [1]) Solve the ODE by determining the appropriate integrating factor.

$$ydx + (2x - ye^y) dy = 0 \quad (13)$$

15. Determine the general solution of the ODE

$$\frac{dy}{dx} + \frac{1}{x}y = x^2y^2 \quad (14)$$

16. (Taken from [2]) Determine the general solution to the ODE

$$dy - (1 + x + y + xy) dx = 0 \quad (15)$$

17. (Taken from [2]) Determine the particular solution to the initial value problem

$$(1 + x) \frac{dy}{dx} + 2y = 3 \cos x \text{ with } y(0) = 1 \quad (16)$$

18. (Taken from [2]) Determine the particular solution to the initial value problem

$$(x^2 + 4) \frac{dy}{dx} + 3xy = x \text{ with } y(0) = 1 \quad (17)$$

19. Solve the first-order ODE

$$x \frac{dy}{dx} + y = x^2 y^2 \quad (18)$$

20. Solve the first-order ODE

$$\frac{dy}{dx} + y = y^2 e^x \quad (19)$$

21. Solve the first-order ODE

$$2x^2 dy - (x^2 + y^2) dx \quad (20)$$

22. Solve the ODE

$$\frac{dy}{dx} + 3x^2 y = 6x^2 \quad (21)$$

23. Solve the ODE

$$\frac{dy}{dx} = \ln(x) (y^4 \ln(x) - y) \quad (22)$$

24. In radio frequency (RF) systems, a power amplifier is used to convert low-power RF signals into high-power RF signals. To power these amplifiers, RL circuits are often used as DC power supplies. Consider an RL circuit where a time-varying voltage source is connected in series with a resistor R and an inductor L . The voltage provided by the source follows an exponential decay function given by $V(t) = V_0 e^{-t}$ where V_0 is the initial voltage at $t = 0$. Using Kirchhoff's Voltage Law (KVL) and the method of integrating factors, find the expression for the current $i(t)$ in the circuit as a function of time. Assume the initial current to be i_0 .

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional. All copyright and trademark rights belong to the original authors and publishers.

References

- [1] G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd ed. Boca Raton, FL: CRC Press, 2016.
- [2] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.